

Customer Churn Analysis: Algorithm & Dataset Specification

Technical Methodology, Process Flow, and Data Dictionary Document

1. DATASET SPECIFICATION

- **Source of Data:** Telecom Customer Churn Repository / Enterprise CRM Database Extract.
- **Number of Records/Samples:** 7,043 unique customer profiles (rows).
- **Data Format & Type:** Structured CSV (`.csv`) tabular format containing numeric, categorical, and binary fields.
- **Training & Testing Split:** 80% Training Set (5,634 samples) and 20% Testing Set (1,409 samples) using Stratified Random Sampling.

Category	Feature / Attribute Name	Data Type	Description
Demographics	<code>gender</code> , <code>SeniorCitizen</code> , <code>Partner</code> , <code>Dependents</code>	Binary / Categorical	Customer demographic profile and household status.
Account Info	<code>tenure</code> , <code>Contract</code> , <code>PaperlessBilling</code> , <code>PaymentMethod</code>	Numeric / Categorical	Active months, contract terms (Month-to-month, 1-yr, 2-yr), billing method.
Services Engaged	<code>PhoneService</code> , <code>InternetService</code> , <code>TechSupport</code> , etc.	Categorical	Active subscriptions and add-on services.
Financial Metrics	<code>MonthlyCharges</code> , <code>TotalCharges</code>	Continuous Numeric	Monthly spend and total cumulative charges.
Target Variable	<code>Churn</code>	Binary Target	Outcome flag: <code>Yes</code> (1 - Churned) vs.

Category	Feature / Attribute Name	Data Type	Description
			No (0 - Retained).

2. ALGORITHM AND SYSTEM ARCHITECTURE

Model & Techniques Applied:

- **Algorithms:** Gradient Boosted Decision Trees (`XGBoost` , `LightGBM`), Random Forest Classifier, and Logistic Regression baseline.
- **Explainability & Evaluation:** `SHAP` values for interpretability; evaluated on ROC-AUC, Recall, Precision, and F1-Score.

Step-by-Step Procedure:

1. **Data Preprocessing:** Handle missing data in `TotalCharges` , convert data types, and scale numeric metrics using `StandardScaler` .
2. **Feature Engineering & Encoding:** One-Hot encode nominal variables and derive interaction ratios (e.g., charge-to-tenure ratio).
3. **Data Partitioning & Balancing:** Split data into 80% train / 20% test via Stratified Sampling; apply `SMOTE` to balance training classes.
4. **Model Training & Optimization:** Train `XGBoost` classifier using 5-fold cross-validation with `GridSearchCV` hyperparameter tuning.
5. **Evaluation & Dashboarding:** Evaluate test set metrics, compute SHAP feature importances, and pass probability risk scores to BI tools.

Process Flow of the System:

[Raw CRM CSV Data] → [Data Cleaning & Imputation] → [Encoding & Feature Scaling] → [80/20 Stratified Split] → [XGBoost Training & Hyperparameter Tuning] → [SHAP Explainability Analysis] → [Risk Tiering] → [BI Dashboard Output]